# Jeremy McMahan

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## Research Interests

Topics — *Algorithms under Uncertainty; Approximation Algorithms; Online, Stochastic and Combinatorial Optimization; Reinforcement Learning; Algorithmic Game Theory; Learning-Augmented Algorithms; Machine Learning*

Summary — Bridging *Approximation Algorithms*, *Reinforcement Learning*, and *Algorithmic Game Theory* to overcome computational barriers facing **tractable sequential decision-making under uncertainty**.

## Education

2018–2025 **PhD in Computer Science**, *University of Wisconsin-Madison*
- Thesis: Safe Multi-Agent Reinforcement Learning in Polynomial Time
- Advisor: Jerry Zhu
- Committee: Yudong Chen, Josiah Hanna, Sandeep Silwal, Qiaomin Xie, Jerry Zhu
- Affiliation: Institute for Foundations of Data Science
- Distinction: LEAP Fellow (NSF Alliance)

2018–2020 **MS in Computer Science**, *University of Wisconsin-Madison*
- Thesis: A D-competitive algorithm for the Multilevel Aggregation Problem with Deadlines
- Advisor: Shuchi Chawla

2016–2018 **BS in Mathematics and Computer Science**, *University of Illinois Urbana-Champaign*
- Thesis: Spectral Graph Isomorphism
- Advisor: Alexandra Kolla
- Distinction: Top graduate in Mathematics and Computer Science

## Publications

**Indicates equal contribution. Authorship order follows venue conventions.*

### Selected Publications

ICML '25 **Polynomial-Time Approximability of Constrained Reinforcement Learning**
Jeremy McMahan
- First polynomial-time $(0, \epsilon)$-bicriteria approximation for general constrained MDPs, the best approximation guarantee possible assuming $P \neq NP$.
- Establishes a unifying framework for tractable constrained decision-making, including almost-sure, chance, and expectation constraints.

NeurIPS '24 **Deterministic Policies for Constrained Reinforcement Learning in Polynomial Time**
Jeremy McMahan
- First FPTAS computing deterministic policies in single-constraint MDPs.
- Resolved the long-standing existence of efficient approximations question for almost-sure, anytime, and deterministic-policy expectation constraints.

ICML '24 **Roping in Uncertainty: Robustness and Regularization in Markov Games**
Jeremy McMahan, Giovanni Artiglio, Qiaomin Xie
- First characterizations and polynomial-time algorithms for equilibria robust to model-uncertainty in Markov games.

AAAI '24 **Optimal Attack and Defense for Reinforcement Learning**
Jeremy McMahan, Young Wu, Xiaojin Zhu, Qiaomin Xie
- First characterizations and polynomial-time algorithms for optimal manipulation attacks and robust equilibrium defenses in Markov games.
- Yields tractable methods for evaluating and ensuring policy robustness to adversarial environmental manipulation.

APPROX '23 **Approximating Pandora's Box with Correlations**
Shuchi Chawla*, Evangelia Gergatsouli*, Jeremy McMahan*, Christos Tzamos*
- First polynomial-time adaptive approximation for Pandora's Box with correlated values; remains the best-known approximation ratio.

arXiv **A D-competitive Algorithm for the Multilevel Aggregation Problem with Deadlines**
Jeremy McMahan
- First $D$-competitive algorithm for multi-level aggregation with deadlines; remains the best-known competitive ratio.

## Additional Refereed Publications

ICML '25 **Anytime-Constrained Equilibria in Polynomial Time**
Jeremy McMahan

RLC '24 **Inception: Efficiently Computable Misinformation Attacks on Markov Games**
Jeremy McMahan, Young Wu, Yudong Chen, Xiaojin Zhu, Qiaomin Xie

ICML '24 **Minimally Modifying a Markov Game to Achieve any Nash Equilibrium and Value**
Young Wu, Jeremy McMahan, Yiding Chen, Yudong Chen, Xiaojin Zhu, Qiaomin Xie

CogSci '24 **Various Misleading Visual Features in Misleading Graphs: Do they truly deceive us?**
Jihyun Rho, Martina A Rau, Shubham Kumar Bharti, Rosanne Luu, Jeremy McMahan, Andrew Wang, Jerry Zhu

AISTATS '24 **Anytime-Constrained Reinforcement Learning**
Jeremy McMahan, Xiaojin Zhu

AAAI '24 **Data Poisoning to Fake a Nash Equilibrium for Markov Games**
Young Wu, Jeremy McMahan, Xiaojin Zhu, Qiaomin Xie

AAAI '23 **Reward Poisoning Attacks on Offline Multi-Agent Reinforcement Learning**
Young Wu, Jeremy McMahan, Xiaojin Zhu, Qiaomin Xie

GAMES '20 **Noble Deceit: Optimizing Social Welfare for Myopic Multi-Armed Bandits**
Ashwin Maran*, Jeremy McMahan*, Nathaniel Sauerberg*

## Research Software

**LLM-Augmented Optimization Agent**, *[GitHub Link]*
An LLM-augmented agent architecture for combinatorial optimization that integrates LLM-generated heuristics with classical algorithm design — achieving empirical efficiency gains from experience while provably retaining worst-case optimality guarantees, demonstrating a principled path to integrating LLM reasoning with formal algorithmic guarantees.

**Constrained Sequential Decision-Making**, *[GitHub Link]*
An algorithmic framework for constrained decision-making, implementing efficient optimization solvers and deep reinforcement learning approaches for anytime-constrained agents. Includes a comprehensive benchmarking suite for evaluating performance tradeoffs. (*AISTATS '24*)

**Robust Multi-Agent Decision-Making**, *[GitHub Link]*
Reference implementations of scalable algorithms for online attacks and defenses in Markov games, enabling empirical evaluation and equilibrium computation for robust multi-agent decision systems. *(AAAI '24)*

**Optimization and Learning for Resource Allocation**, *[GitHub Link]*
A large-scale evaluation framework demonstrating the tradeoffs between classical approximation algorithms and deep-learning heuristics on resource allocation tasks, providing empirical grounding for theory-first algorithm design in AI systems.

## Selected Talks

**From Knapsacks to Self Driving: FPTAS Recipes for Constrained Reinforcement Learning**
- UMD College Park (MARL Seminar) *March 2025*
- UW-Madison (Theory of Computing Seminar) *November 2024*

**Optimal Attack and Defense for Reinforcement Learning**
- UW-Madison (Theory of Computing Seminar) *February 2025*
- UW-Madison (Computer Vision Roundtable) *January 2024*

## Teaching Experience

### Primary Instructor

Summer 2022 — **CS 577: Introduction to Algorithms**, *University of Wisconsin-Madison*
- Responsible for curriculum design, lecture delivery, course structure, and student evaluation for a class of 100 students.
- Designed original lecture slides, lecture notes, lecture videos, and problem sets focusing on the design of efficient algorithms and rigorous proofs of correctness.
- Managed one TA and grader.

### Head Teaching Assistant

Spring 2021, 2022 — **CS 540: Introduction to Artificial Intelligence**, *University of Wisconsin-Madison*
- Coordinated the administrative and grading workflow for a large-scale course with 500 students, 8 TAs, and over 20 graders and peer mentors.
- Developed standardized grading rubrics and course structures in addition to the standard teaching assistant duties.
- Instructors: Ilias Diakonikolas, Sharon Li, Fred Sala, and Jerry Zhu

### Guest Lecturer

Spring 2025 — **CS 839: Game Theory, Optimization & Learning**, *University of Wisconsin-Madison*
- Designed and presented two lectures on the theory of multi-agent reinforcement learning.
- Instructor: Manolis Vlatakis

### Graduate Teaching Assistant

Summer 2021, 2022 — **New Horizons in Theoretical Computer Science Summer School**, *ACM SIGACT*
- Designed lectures spanning complexity theory basics, interactive proofs, and Markov-chain Monte Carlo methods.
- Instructors: Antonio Blanca, Nicole Immorlica, Yael Kalai, Jelani Nelson, and Ashia Wilson

Fall 2020, 2021 — **CS 540: Introduction to Artificial Intelligence**, *University of Wisconsin-Madison*
Instructors: Anthony Gitter, Josiah Hanna, Yin Li, Yingyu Liang, and Daifeng Wang

Spring 2019, 2020 — **CS 577: Introduction to Algorithms**, *University of Wisconsin-Madison*
Instructors: Shuchi Chawla, Dieter van Melkebeek, and Christos Tzamos

Fall 2019 — **CS 301: Introduction to Data Science**, *University of Wisconsin-Madison*
Instructor: Tyler Caraza-Harter

Fall 2018 — **CS 524: Introduction to Optimization**, *University of Wisconsin-Madison*
Instructor: Steve Wright

## Awards and Honors

2021–2025 — **LEAP Fellow**, *CMD-IT / NSF Alliance*
Selected for the NSF-funded Leadership, Excellence, Access, and Pathways (LEAP) Alliance fellowship.

2023 **Graduate Instructor Award**, *University of Wisconsin-Madison*
Recognition for excellent performance as a primary instructor.

2019–2022 **The "Golden Brick" Teaching Award**, *University of Wisconsin-Madison*
A student/instructor-voted honor for most effective teaching. Won consecutively for every year served as a TA (4 years).

2018–2019 **CS Departmental Scholarship**, *University of Wisconsin-Madison*
Merit-based scholarship awarded to incoming PhD students.

2018 **Most Outstanding Major Award**, *University of Illinois Urbana-Champaign*
Awarded to the #1-ranked student in Mathematics and Computer Science.

## Professional Service

2024–2025 **Conference Reviewer**
NeurIPS (2024, 2025), ICML (2024, 2025), AISTATS (2025), RLC (2024)

2023–2025 **Co-Organizer**, *Machine Learning Lunch Meetings (MLLM) Seminar*, *University of Wisconsin-Madison*
Organized weekly research seminars for the ML community, which included managing speaker invitations, scheduling, and logistics for internal and external presenters.

2021–2022 **Planning Committee & Teaching Assistant**, *ACM SIGACT*, *New Horizons in Theoretical Computer Science Summer School*
A week-long summer school aimed at inspiring undergraduates from underrepresented backgrounds to pursue careers in TCS.
- Served on the planning committee.
- Mentored students during technical sessions and social events.
- Developed and presented lessons on selected TCS topics.

## Technical Skills

Languages: Python, Julia, C/C++, Mathematica, SageMath, LaTeX, Bash

Libraries: Gymnasium, LangGraph, PyTorch, OpenAI API, Gurobi, SciPy, NumPy, Pandas